

Jude Rifai

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EDUCATION

Georgia Institute of Technology

Atlanta, GA

BS in Computer Science & BS in Mechanical Engineering; GPA: 3.96/4.0

May 2028

Relevant Coursework: Data Structures and Algorithms, Artificial Intelligence, Robotics and Perception, Machine Learning, Computer Architecture, Database Systems

TECHNICAL SKILLS

Languages/Platforms: C/C++, Python, Java, JavaScript, SQL (Lite + Postgres), Linux, Docker, Git

Frameworks: React.js, Next.js, ROS, MuJoCo, FastAPI, Flask, Node.js, Express.js, Django

Libraries: PyTorch, NumPy, pandas, Scikit-learn, OpenCV, TensorFlow, Transformers, Matplotlib

EXPERIENCE

HALab Undergraduate Researcher

Jan. 2026 – Present

Georgia Institute of Technology

Atlanta, GA

- Engineered and benchmarked **3+** VLA architectures (**GR00T**, **ACP**, **RDT**) for robotic manipulation
- Trained baseline policies on **custom teleoperation data** spanning **4 manipulation tasks**
- Boosted policy success rate by **30%** by fusing **wrist and overhead** camera views

Yellow Jacket Space Program Avionics Member

Jan. 2025 – Present

Georgia Institute of Technology

Atlanta, GA

- Developed embedded **C/C++** firmware streaming **4 sensor channels** (IMU, pressure, temp) at **100 Hz**
- Drove communication faults down **80%** through automated **PCB validation** scripts

Software Engineering Intern

Nov. 2023 – Apr. 2024

United Road Services

Detroit, MI

- Automated a **Python** pipeline optimizing **3+ metrics** (cost, performance, reliability) across **13 states**
- Shipped an internal **Django + PostgreSQL** dashboard visualizing fleet data for **100+ EVs**

PROJECTS

SleepTracker | *Python, PyTorch, CNN14, BiLSTM, Audio Processing*

July 2026

- Developed **3+ ML models** to classify sleep stages from **MFCC** and **mel spectrogram** audio
- Fine-tuned an **80M-parameter** CNN14 + BiLSTM model for **5-class** sleep staging
- Reached **80% accuracy** on **90-minute** polysomnography sequences

RobotVision | *C++, Python, ROS, GroundedSAM, FoundationPose*

May 2026

- Benchmarked **2** pose-estimation models (**GroundedSAM**, **FoundationPose**) for **6D** humanoid pose estimation
- Evaluated **speed, accuracy (ADD-S)**, and **occlusion robustness** across **5 test scenes**

TempoSchedular | *React, Node.js, Express, SQLite*

Nov. 2025

- Architected a full-stack weekly scheduler with a **React** frontend and **10+ endpoint Node/Express API**
- Built **real-time conflict detection** with **OpenAI**-powered scheduling, tested by **30 users**

LEADERSHIP & COMMUNITY INVOLVEMENT

President and Founder, Kappa Theta Pi

Aug. 2026 – Present

Georgia Institute of Technology

Atlanta, GA

- Founded and now lead a **20-member** class launching **Kappa Theta Pi's** GT chapter
- Directed a **\$2,000+** budget across **15+** workshops, speaker panels, and technical events

Teaching Assistant, Data Structures & Algorithms

May 2026 – Present

Georgia Institute of Technology

Atlanta, GA

- Mentored **1000+ students/semester** via office hours, exam prep, and recruiting events
- Taught **75+ minute** weekly recitations on trees, graphs, and DP collaborating with a **20-person** TA team